01NAEX - Lecture 03
Randomized Blocks and Factorial Designs

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Introduction to Blocking, Nuisance Factors, and Factorial Design

D. C. Montgomery, Design and Analysis of Experiments: Sections 4.1 and 5.1–5.6

- The randomized complete block design (RCBD): model, ANOVA, relative efficiency, comparisons of treatment means, parameter estimates, random blocks.
- Sample size for the RCBD (OC curves, noncentral F).
- General principles of factorial experiments: main effects, interaction, why not one factor at a time.
- The two-factor factorial with fixed effects: ANOVA, model checking, comparisons of means, simple effects.
- One observation per cell: Tukey's test for nonadditivity.
- Sample size for factorial designs.
- Quantitative factors: response curves and response surfaces.
- Factorials with more than two factors; blocking in a factorial design.
- Other blocking scenarios (Latin squares, Graeco-Latin squares) and incomplete block designs will be covered in the next lecture.

Introduction to Blocking and Nuisance Factors

Blocking is a technique for dealing with nuisance factors

- A nuisance factor is a factor that probably has some effect on the response, but it is of no interest to the experimenter however, the variability it transmits to the response needs to be minimized.
- Typical nuisance factors include batches of raw material, operators, pieces of test equipment, time (shifts, days, etc.), different experimental units.
- Failure to block is a common flaw in the design of an experiment and data analysis.

If the nuisance factor is known and controllable, we use blocking. Other cases:

- If the nuisance factor is known and uncontrollable, sometimes we can use the analysis of covariance to remove the effect of the nuisance factor from the analysis.
- If the nuisance factor is unknown and uncontrollable, we hope that randomization balances out its impact across the experiment.
- Sometimes several sources of variability are combined in a block, so the block becomes an aggregate variable.

When to Block vs When to Just Randomize

Nuisance vs noise

- **Nuisance factor** = known source of systematic variability.
- **Noise** = unknown, many tiny influences you can't name.

Decision guide

- **Known & assignable** (day/operator/lot) $\Rightarrow$ **block** it.
- **Unknown/weak** $\Rightarrow$ rely on **randomization**, check residuals vs run.
- **Trade-off:** Blocking uses $b-1$ **df**; does it noticeably **reduce** MS_E?

Practical checklist

- Define blocks so each block is **as homogeneous as possible**.
- **Randomize within each block**. (Pre-print run sheets.)
- If the block effect looks tiny, a CRD would have done as well (see relative efficiency).

Rule of thumb: If **Monday** is a nuisance factor, block it.

Q: In a thermal test, which would you block - **oven**, **operator**, or **day**? Why?

Model (no treatment-by-block interaction):
$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$, $i = 1, \ldots, a$ (treatments), $j = 1, \ldots, b$ (blocks), $\sum \tau_i = \sum \beta_j = 0$.

Source	df	SS	MS / Test
Treatments (τ)	$a-1$	SS_T	$MS_T = SS_T/df_T$
Blocks (β)	$b-1$	SS_B	$MS_B = SS_B/df_B$
Error	$(a-1)(b-1)$	SS_E	$MS_E = SS_E/df_E$

Tests: $F_T = MS_T/MS_E$ (treatment), $F_B = MS_B/MS_E$ (block).
Note: With one observation per cell the treatment-by-block interaction cannot be estimated; Tukey's test for nonadditivity (later today) checks the additivity assumption.

What Blocking Buys You

Model: $y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$, $\varepsilon_{ij} \sim (0, \sigma^2)$, blocks variability $\mathrm{Var}(\beta_j) = \sigma_\beta^2$.

Without block (CRD): block shifts leak into error $\Rightarrow$ error variance $\approx \sigma^2 + \sigma_\beta^2$.
With block (RCBD): the block mean is removed $\Rightarrow$ error variance stays σ^2.

Gain in signal-to-noise:

$$F = \frac{MS_\tau}{MS_E} \text{ increases with: Gain} \approx \frac{\sigma^2 + \sigma_\beta^2}{\sigma^2} = 1 + \frac{\sigma_\beta^2}{\sigma^2}.$$

Example: If measurement noise $\sigma = 1$ and day-to-day shifts $\sigma_\beta = 2$,

$$\text{Gain} \approx 1 + \frac{4}{1} = 5\times \text{ (much larger } F \text{ with blocks).}$$

- Always **randomize within blocks** and check residuals vs run order.

Q: If σ_β^2 is nearly zero, what is the cost/benefit of blocking? *Hint:* You spend df but gain little precision.

The Hardness Testing Example

- We wish to determine whether different tips produce different (mean) hardness reading on a tester machine.
- 4 different types of tips: conical, ball, cylinder, and cube.
- The test coupons (experimental unit) are a source of nuisance variability.
- Experimental error measures both random error and variability between coupons.
- To conduct this experiment as a RCBD, assign all 4 tips to each coupon.
- Each coupon is called a **block**; that is, it is a more homogeneous experimental unit on which to test the tips.
- Variability between blocks can be large, variability within a block should be relatively small.

In general, a block is a specific level of the nuisance factor. A complete replicate of the basic experiment is conducted in each block and all runs within a block are randomized.

The Hardness Testing Example

Randomized Complete Block Design for the Hardness Testing Experiment

Test Coupon (Block)			
1	**2**	**3**	**4**
Tip 3	Tip 3	Tip 2	Tip 1
Tip 1	Tip 4	Tip 1	Tip 4
Tip 4	Tip 2	Tip 3	Tip 2
Tip 2	Tip 1	Tip 4	Tip 3

In completely randomized single-factor design, 16 different metal test coupons would be required, one for each run in the design.

In RCBD, the variability between coupons from the experimental error is removed.

Only randomization of treatments (Tips) is within the blocks, i.e. blocks represent a restriction on randomization.

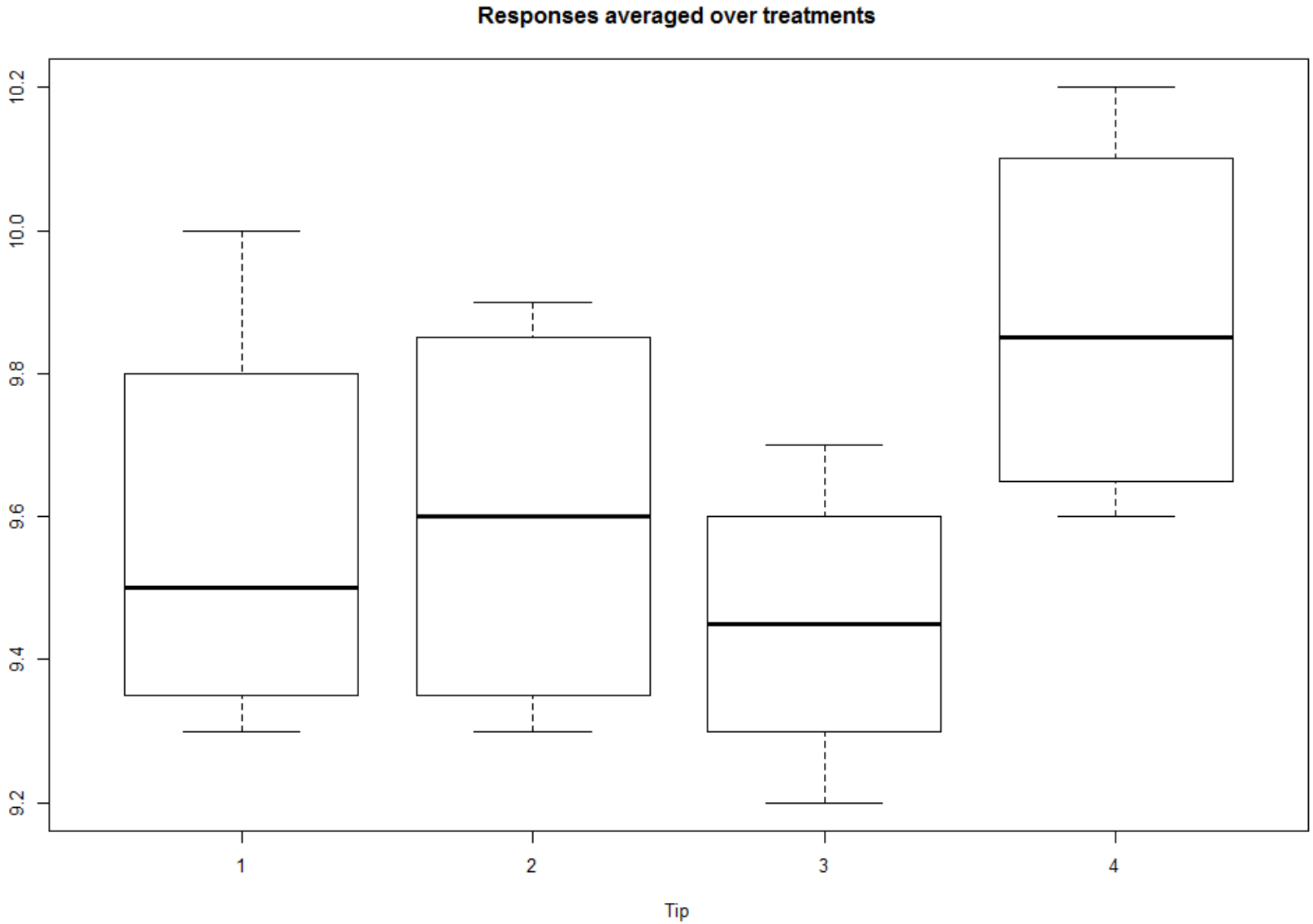
The Hardness Testing Example

The two-way structure of the experiment (Tips x Blocks)

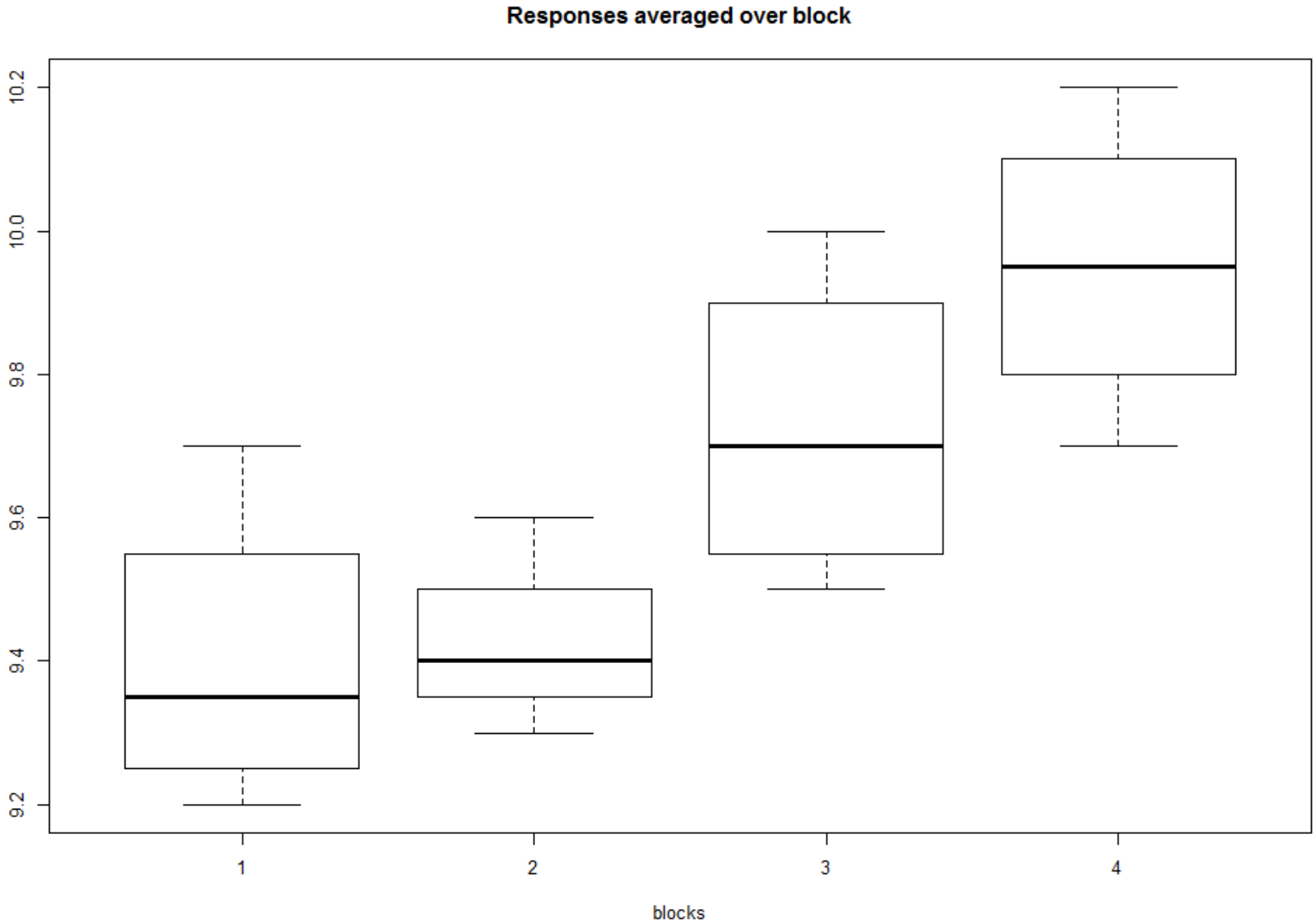
Tip	Test Coupon (Block)			
Number	1	2	3	4
1	9.3	9.4	9.6	10.0
2	9.4	9.3	9.8	9.9
3	9.2	9.4	9.5	9.7
4	9.7	9.6	10.0	10.2

- We are interested in testing the equality of treatment means (again).
- We have to remove the variability associated with the nuisance factor (the blocks).
- Analysis is identical to the two-factor factorial design without replication.

The Hardness Testing Example

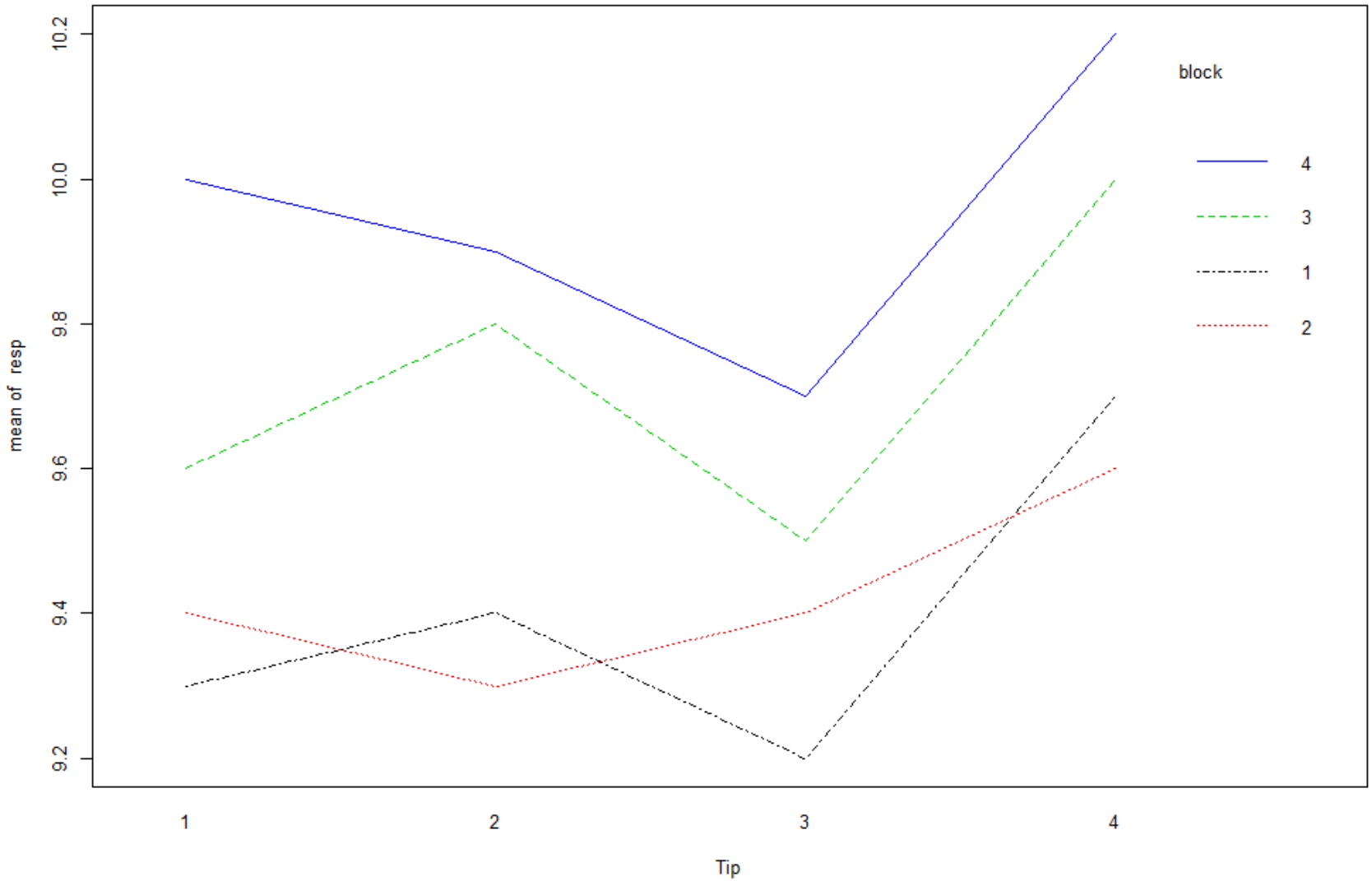


The Hardness Testing Example



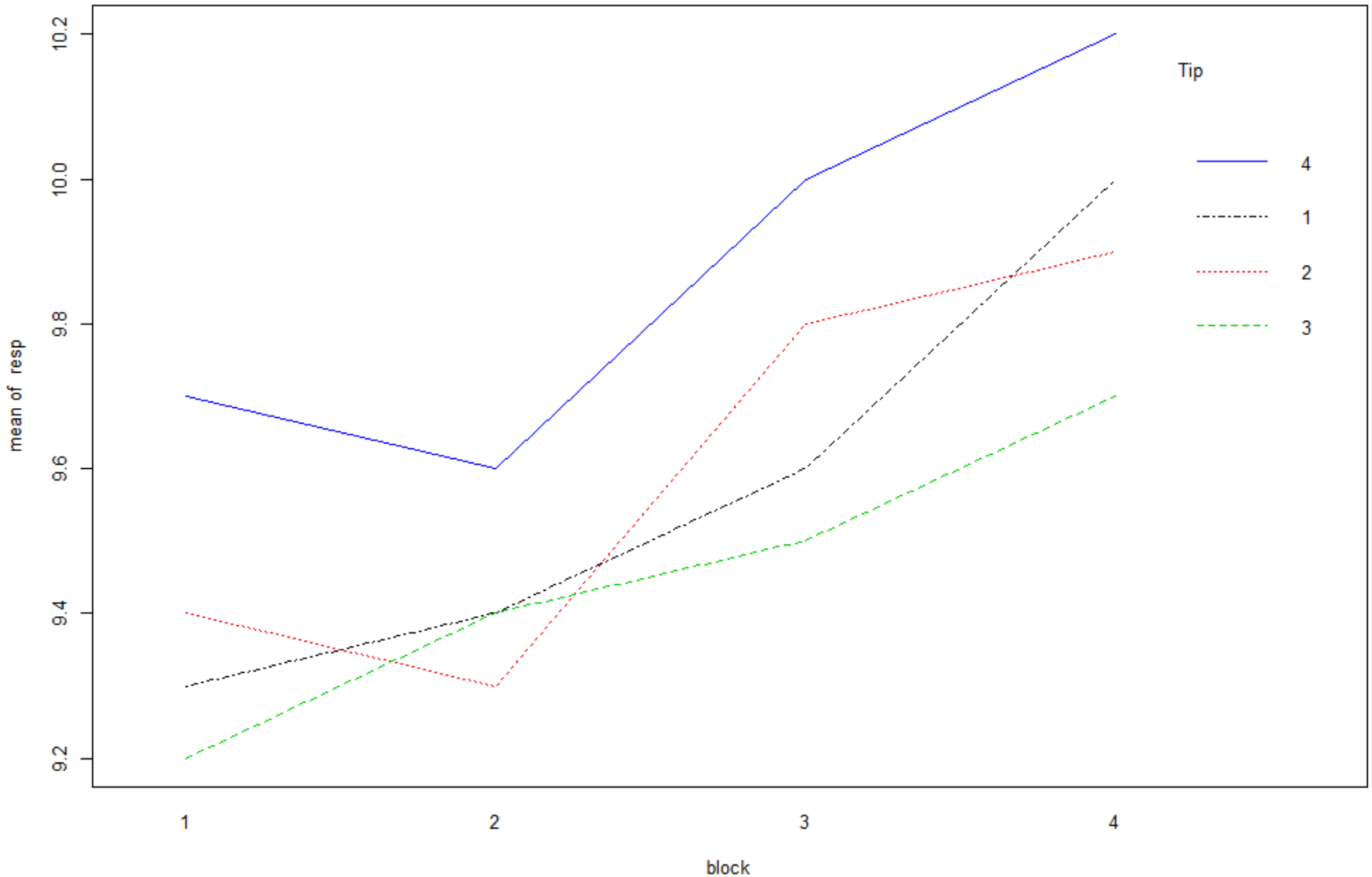
The Hardness Testing Example

Interaction plot over Tips.



The Hardness Testing Example

Interaction plot over Blocks.



ANOVA for the RCBD

Effects model for the RCBD:

$$y_{ij} = \mu_{ij} + \epsilon_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij} \quad i = 1, 2, \ldots, a \quad j = 1, 2, \ldots, b,$$

where μ is an overall mean, τ_i is the effect of the ith treatment, β_j is the effect of the jth block and $\epsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ random error term.

Analysis of Variance for a Randomized Complete Block Design

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Treatments	$SS_{\text{Treatments}}$	$a - 1$	$\frac{SS_{\text{Treatments}}}{a - 1}$	$\frac{MS_{\text{Treatments}}}{MS_E}$
Blocks	SS_{Blocks}	$b - 1$	$\frac{SS_{\text{Blocks}}}{b - 1}$	
Error	SS_E	$(a - 1)(b - 1)$	$\frac{SS_E}{(a - 1)(b - 1)}$	
Total	SS_T	$N - 1$		

$$SS_E = SS_T - SS_{Treatments} - SS_{Blocks}$$

$$= \sum_{i=1}^{a}\sum_{j=1}^{b}(y_{ij} - \bar{y}_{..})^2 - b\sum_{i=1}^{a}(\bar{y}_{i.} - \bar{y}_{..})^2 - a\sum_{j=1}^{b}(\bar{y}_{.j} - \bar{y}_{..})^2$$

The Hardness Testing Example

```
y = [[9.3, 9.4, 9.6, 10.0], [9.4, 9.3, 9.8, 9.9],
     [9.2, 9.4, 9.5, 9.7], [9.7, 9.6, 10.0, 10.2]]
rows = [(t + 1, c + 1, y[t][c]) for t in range(4)
        for c in range(4)]
hard = pd.DataFrame(rows, columns=["Tip", "Coupon", "y"])
rcbd = smf.ols("y ~ C(Tip) + C(Coupon)", data=hard).fit()
sm.stats.anova_lm(rcbd, typ=2)
             sum_sq   df         F     PR(>F)
C(Tip)        0.385  3.0    14.438   0.000871
C(Coupon)     0.825  3.0    30.938   0.000045
Residual      0.080  9.0
```

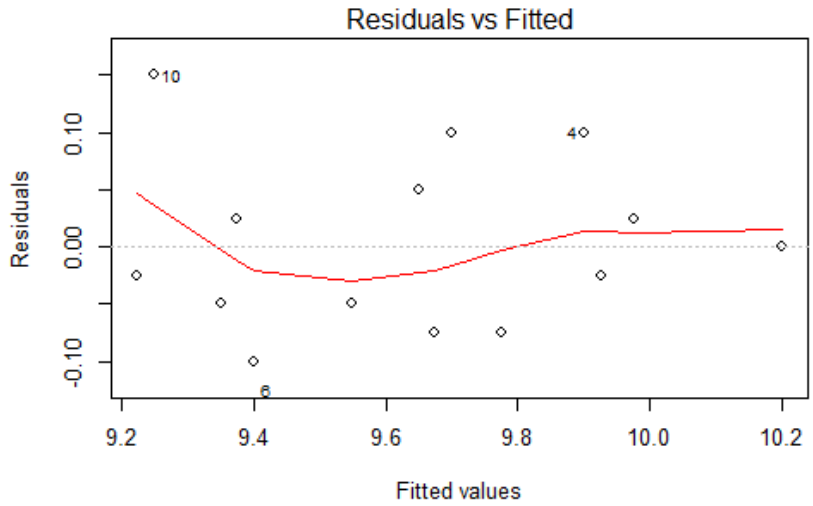
The same data analysed as if they came from a completely randomized design:

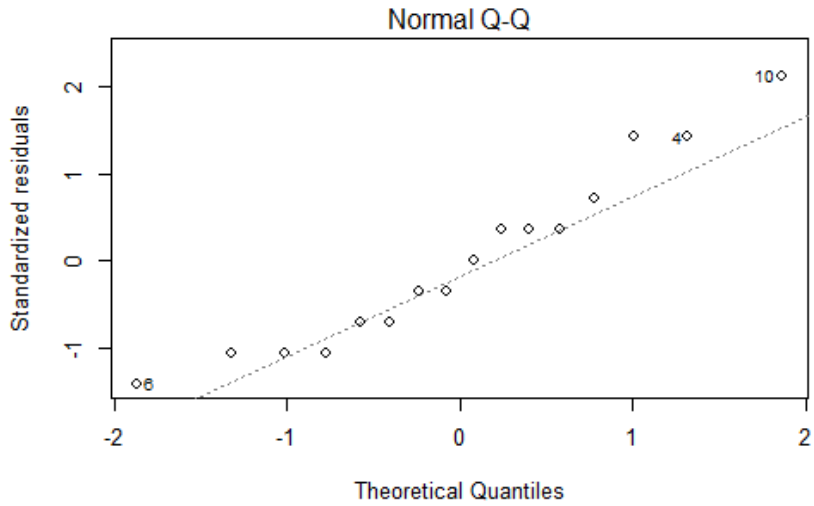
```
crd = smf.ols("y ~ C(Tip)", data=hard).fit()
sm.stats.anova_lm(crd, typ=2)
            sum_sq    df         F    PR(>F)
C(Tip)       0.385   3.0    1.702    0.2196
Residual     0.905  12.0
```

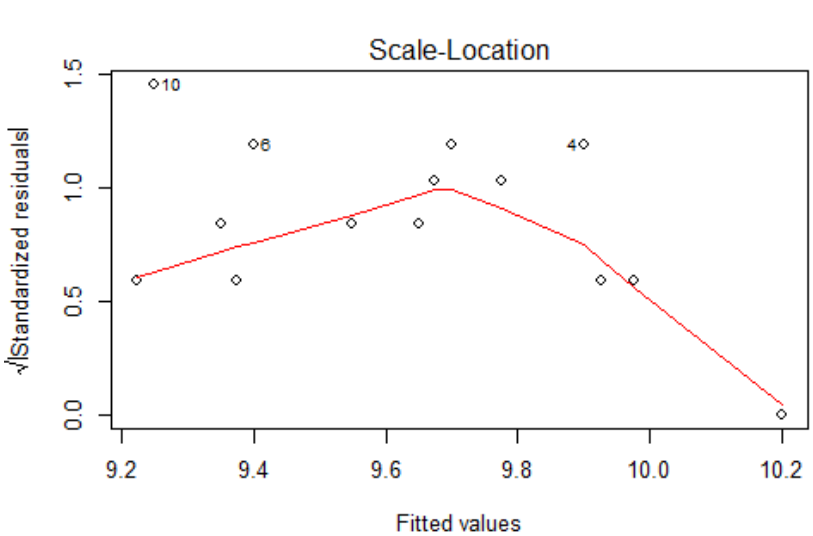
The residual mean square grows from 0.0089 to 0.0754 (about 8.5 times): all the variability between coupons is now in the error term and the tip effect is no longer detected. This naive ratio overstates the gain from blocking; the relative efficiency (two slides ahead) gives the proper comparison.

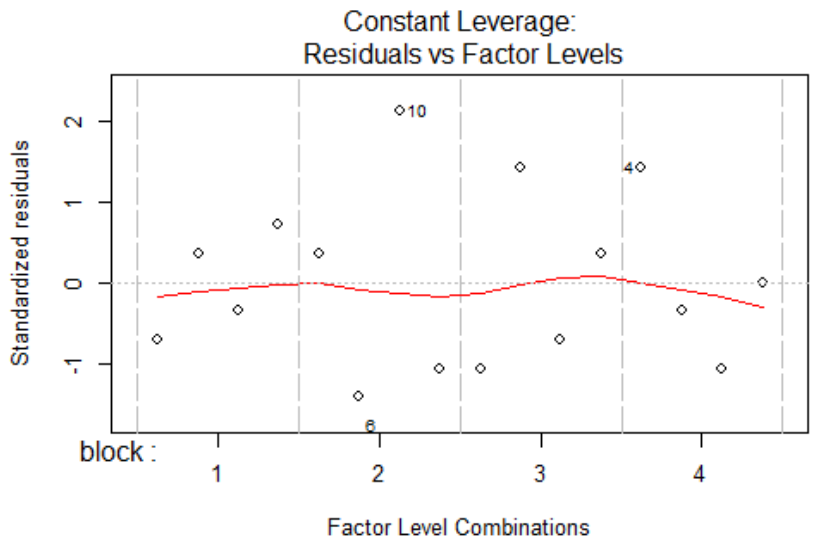
The Hardness Testing Example

Model adequacy checking:









Relative Efficiency of the RCBD

Montgomery, Supplemental Material S4.1

How many more observations would a completely randomized design (CRD) need to estimate the treatment means as precisely as the RCBD?

$$R = \frac{(df_b + 1)(df_r + 3)}{(df_b + 3)(df_r + 1)} \frac{\hat{\sigma}_r^2}{\hat{\sigma}_b^2}, \qquad \hat{\sigma}_r^2 = \frac{(b-1)MS_{Blocks} + b(a-1)MS_E}{ab - 1},$$

where $\hat{\sigma}_r^2$ estimates the error variance of a CRD with the same units, $\hat{\sigma}_b^2 = MS_E$, and $df_b = (a-1)(b-1)$, $df_r = a(b-1)$ are the error df of the RCBD and the CRD.

Hardness example, from the models `rcbd` and `crd` fitted above:

```
a = b = 4
s2b = rcbd.mse_resid                                  # 0.00889
s2r = (crd.ssr + (a - 1) * s2b) / (a * b - 1)         # 0.0621
dfb, dfr = rcbd.df_resid, crd.df_resid                # 9, 12
R = (dfb + 1) * (dfr + 3) / ((dfb + 3) * (dfr + 1)) * s2r / s2b
# 6.72   (s2r / s2b = 6.99)
```

`s2r` is $\hat{\sigma}_r^2$, since $0.905 + 3(0.00889) = (b-1)MS_{Blocks} + b(a-1)MS_E$. Not `crd.mse_resid` $= 0.0754$: each tip met every coupon once, so it puts all coupon variability on 12 df instead of 15. A CRD would need about 6.7 times as many observations per tip; blocking was clearly worth its 3 df.

Q: What value of R would you expect if the coupons were all alike?

The Hardness Testing Example - Which Tips Differ?

Multiple comparisons use MS_E and df_E of the **RCBD model** ($n = b = 4$ observations per tip):

```
mse, dfe = rcbd.mse_resid, rcbd.df_resid     # 0.00889, 9
means = hard.groupby("Tip").y.mean()   # 9.575 9.600 9.450 9.875
lsd = stats.t.ppf(0.975, dfe) * np.sqrt(2 * mse / 4)      # 0.151
q = stats.studentized_range.ppf(0.95, 4, dfe)             # 4.415
hsd = q * np.sqrt(mse / 4)                                # 0.208
```

pair	diff	|diff| > LSD	|diff| > HSD
2-1	0.025	no	no
3-1	-0.125	no	no
4-1	0.300	yes	yes
3-2	-0.150	no	no
4-2	0.275	yes	yes
4-3	0.425	yes	yes

Tip 4 gives higher hardness readings than tips 1, 2 and 3; tips 1, 2 and 3 do not differ.

Caution: `pairwise_tukeyhsd(hard.y, hard.Tip)` fits a one-way model and uses $MS_E = 0.0754$ with 12 df, i.e. it ignores the blocks and finds no difference at all.

Estimating the Model Parameters; Random Blocks

Montgomery, Sections 4.1.3 and 4.1.4

Least squares estimates for fixed effects ($\sum \tau_i = \sum \beta_j = 0$):

$$\hat{\mu} = \bar{y}_{..}, \qquad \hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}, \qquad \hat{\beta}_j = \bar{y}_{.j} - \bar{y}_{..}, \qquad \hat{y}_{ij} = \bar{y}_{i.} + \bar{y}_{.j} - \bar{y}_{..}.$$

Random blocks. If the blocks are a random sample from a population (subjects, batches, days), $\beta_j \sim N(0, \sigma^2_\beta)$ and

$$E(MS_{Blocks}) = \sigma^2 + a\sigma^2_\beta, \qquad E(MS_E) = \sigma^2 \quad \Longrightarrow \quad \hat{\sigma}^2_\beta = \frac{MS_{Blocks} - MS_E}{a}.$$

The test for treatments is unchanged, $F_0 = MS_{Treatments}/MS_E$. A negative estimate is set to zero.

```
g = hard.y.mean()                                  # 9.625
tau = hard.groupby("Tip").y.mean() - g
# -0.050 -0.025 -0.175 0.250
beta = hard.groupby("Coupon").y.mean() - g
# -0.225 -0.200 0.100 0.325
ms_blocks = 0.825 / 3                              # 0.275
s2_beta = (ms_blocks - rcbd.mse_resid) / 4         # 0.0665
```

Coupon-to-coupon variance $\hat{\sigma}^2_\beta = 0.0665$ is about 7.5 times the error variance $\hat{\sigma}^2 = 0.0089$.

The Hardness Testing Example - Number of Blocks

How many blocks do we need to detect a true maximum difference in mean readings of $D = 0.4$ with high probability, if the standard deviation is $\sigma = 0.1$? Least favourable configuration (two means D apart, the others in the middle):

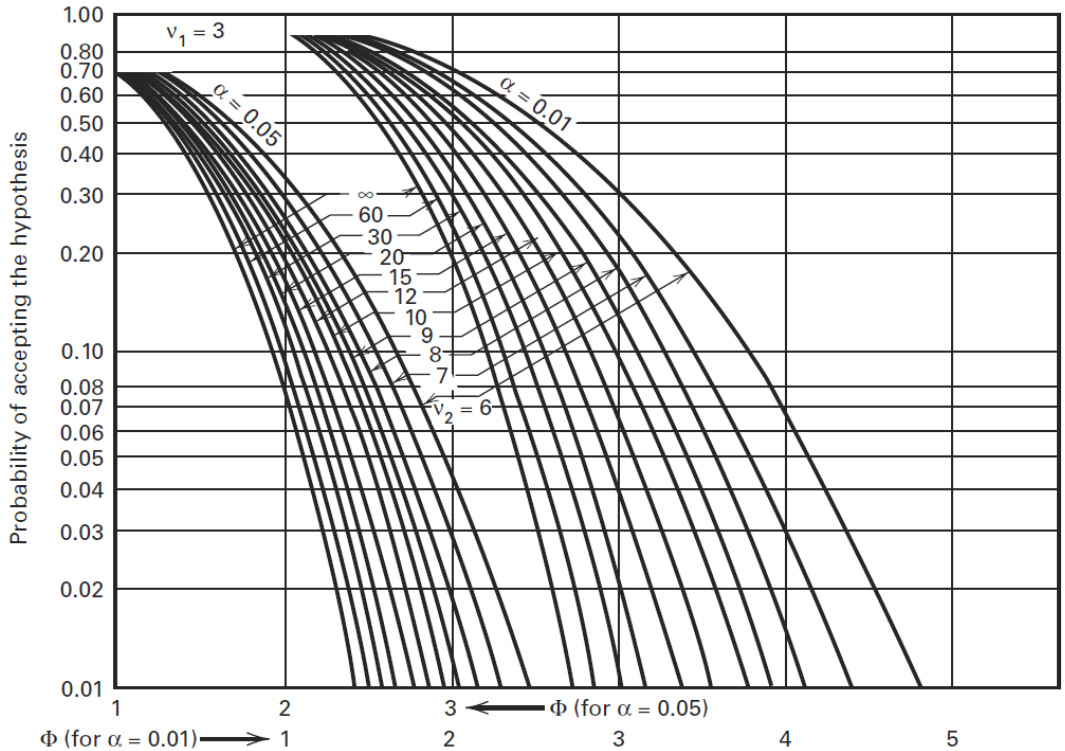
$$\Phi^2 = \frac{bD^2}{2a\sigma^2}, \qquad \lambda = a\Phi^2 = \frac{bD^2}{2\sigma^2},$$

$$F_0 = \frac{MS_{Treatments}}{MS_E} \sim F_{a-1,\,(a-1)(b-1)}(\lambda) \text{ under } H_1.$$

```
a, D, sigma = 4, 0.4, 0.1
for b in range(3, 9):
    lam = b * D**2 / (2 * sigma**2)          # = a * Phi^2
    df1, df2 = a - 1, (a - 1) * (b - 1)
    fcrit = stats.f.ppf(0.95, df1, df2)
    power = stats.ncf.sf(fcrit, df1, df2, lam)
```

b	3	4	5	6	7	8
Phi	2.449	2.828	3.162	3.464	3.742	4.000
lambda	24	32	40	48	56	64
df2	6	9	12	15	18	21
power	0.8461	0.9757	0.9972	0.9997	1.0000	1.0000

OC Curves for the Fixed-Effects Model ANOVA



$\nu_1 = a - 1 = 3$ numerator df, $\nu_2 = (a-1)(b-1)$ denominator df. Reading the chart at Φ gives β; the exact power from the noncentral F:

```
b          3       4       5       6       7       8
Phi    2.449   2.828   3.162   3.464   3.742   4.000
df2        6       9      12      15      18      21
power 0.8461  0.9757  0.9972  0.9997  1.0000  1.0000
```

Already $b = 4$ coupons give power about 0.98.

Power Planning - What Are We Guaranteeing?

Goal: Choose n so the test rejects with probability $\geq 1 - \beta$ when *meaningful* differences exist (at level α).

Two common settings

- **RCBD (Treatment + Block):** Power the **treatment** main effect A. Factor B is a nuisance that reduces noise.
- **Two-treatment factorial:** Power A, B, and **interaction** $A \times B$.

Detectable size(s): Specify minimally important differences:

$$D_A, \quad D_B, \quad D_{AB} \text{ (difference-of-differences).}$$

Q: Is your study

- confirmatory (stricter Family Wise Error Rate (FWER) control)?
- exploratory (report CIs with less stringent False Discovery Rate (FDR) control)?

RCBD (Treatment + Block) & Operating Characteristic (OC)

Model (balanced, n per cell):

$$y_{ijr} = \mu + \tau_i + \beta_j + \varepsilon_{ijr}, \quad i = 1..a,\ j = 1..b,\ r = 1..n, \quad \varepsilon_{ijr} \sim N(0, \sigma^2),$$

$\sum_i \tau_i = \sum_j \beta_j = 0.$

Test of interest (treatments): $H_0 : \tau_1 = \cdots = \tau_a = 0$ with

$$F_0 = \frac{MS_A}{MS_E}, \qquad \mathrm{df}_A = a - 1, \quad \mathrm{df}_E = \begin{cases} (a-1)(b-1), & n = 1 \\ abn - a - b + 1, & n > 1 \end{cases}$$

Under H_1: $F_0 \sim F_{\text{num df, den df}}(\lambda)$ (noncentral F).

Power at level α:

$$1 - \beta \;=\; \Pr\Big(F_{\text{num df, den df}}(\lambda) \;>\; F_{\alpha;\,\text{num df, den df}} \Big).$$

Scaled distance (OC parameter): We use

$$\Phi^2 \equiv \frac{\lambda}{\text{num df} + 1} \quad (\text{num df} + 1 = \text{number of means compared}),$$

so OC curves (Montgomery, Appendix Chart V) are indexed by $(\alpha, \mathrm{df}_{\text{num}}, \mathrm{df}_{\text{den}}, \Phi)$.
Practical view: Bigger Φ^2 (or λ) $\Rightarrow$ larger separation from H_0 $\Rightarrow$ higher power.

Noncentrality for Treatment (RCBD)

Let $\mu_{i\cdot}$ be the block-averaged mean for treatment i and $\bar{\mu}$ the grand mean. For balanced RCBD:

$$\lambda_A = \frac{nb}{\sigma^2}\sum_{i=1}^{a}(\mu_{i\cdot} - \bar{\mu})^2, \qquad \Phi_A^2 = \frac{\lambda_A}{a}.$$

Design target (pairwise): Detect *any* pairwise difference $\geq D_A$.
Least favorable configuration (hardest to detect):

$$\mu_{1\cdot} = +D_A/2, \quad \mu_{2\cdot} = -D_A/2, \quad \mu_{3\cdot} = \cdots = \mu_{a\cdot} = 0 \Rightarrow \sum(\mu_{i\cdot} - \bar{\mu})^2 = D_A^2/2.$$

Therefore

$$\boxed{\Phi_A^2 = \frac{nb\,D_A^2}{2a\sigma^2}}, \qquad \lambda_A = a\,\Phi_A^2 = \frac{nb\,D_A^2}{2\sigma^2}.$$

Check: If $b = 1$, this reduces to the one-way formula.

Choosing n in RCBD (From OC Curves)

Given α, target power, a, b, and pilot $\hat{\sigma}^2$:

Recipe

1. From OC tables (or software), find $\Phi^2_{A,\star}$ that achieves the power with $\mathrm{df}_A = a - 1$, df_E as per design.
2. Solve for n from $\Phi^2_A = \dfrac{nb\,D_A^2}{2a\,\hat{\sigma}^2}$:

$$\boxed{n \;=\; \frac{2a\,\hat{\sigma}^2}{b\,D_A^2}\;\Phi^2_{A,\star}} \quad \text{(round up)}.$$

Heuristics: Halve $\hat{\sigma}^2 \Rightarrow$ halve n; double $D_A \Rightarrow$ quarter n; double $b \Rightarrow$ halve n.

Note: $\Phi_{A,\star}$ depends on df_E, which depends on n: iterate, or compute the power directly from the noncentral F for a few values of n (next slides).

Introduction to Factorial Design

- Factorial Designs are most efficient methods for experiments where we study the effects of two or more factors.
- The effect of the primary factor of interest in the experiment is called **main effect**.
- If the difference in response between the levels of one factor is not the same at all levels of other factors, the **interaction** occurs.

Factorial Basics (2×2) - Main Effects & Interaction

Cell means

	B^-	B^+
A^-	μ_{--}	μ_{-+}
A^+	μ_{+-}	μ_{++}

Main effect of A:
$\frac{1}{2}(\mu_{+-} + \mu_{++}) - \frac{1}{2}(\mu_{--} + \mu_{-+})$

Main effect of B:
$\frac{1}{2}(\mu_{-+} + \mu_{++}) - \frac{1}{2}(\mu_{--} + \mu_{+-})$

Interaction AB:
$\frac{1}{2}\left[(\mu_{++} - \mu_{+-}) - (\mu_{-+} - \mu_{--})\right]$
(half the difference between the effect of B at A^+ and at A^-).

Reading the interaction plot

- *Parallel lines* $\Rightarrow$ no interaction (additive effects).
- *Non-parallel lines* $\Rightarrow$ interaction; *crossing lines* $\Rightarrow$ the sign of one effect depends on the other factor.
- With a significant interaction, main effects can be misleading: look at the **simple effects** (effect of A at a fixed level of B).

Q: Can a factor have zero main effect and still matter?

Why Factorials? (vs One-Factor-at-a-Time)

Montgomery, Section 5.2

One-factor-at-a-time (OFAT) with two factors: run A^-B^-, A^+B^-, A^-B^+; estimate the effect of A from $A^+B^- - A^-B^-$ and of B from $A^-B^+ - A^-B^-$.

Factorial: run all four combinations. Each effect is a difference of two averages of two observations.

- Same precision: OFAT needs 6 runs (each comparison replicated), the factorial needs 4; relative efficiency $6/4 = 1.5$, growing with the number of factors.
- Every run is used for every effect (*hidden replication*).
- The factorial estimates the **interaction**; OFAT cannot, and with an interaction its conclusions depend on the arbitrarily chosen starting point.
- Effects are estimated over a range of conditions of the other factors, so the conclusions are valid over that range.

Q: In Team Project 1 you will vary string length, mass and angle of a pendulum. Why should you not change one factor at a time?

The Battery Design Example

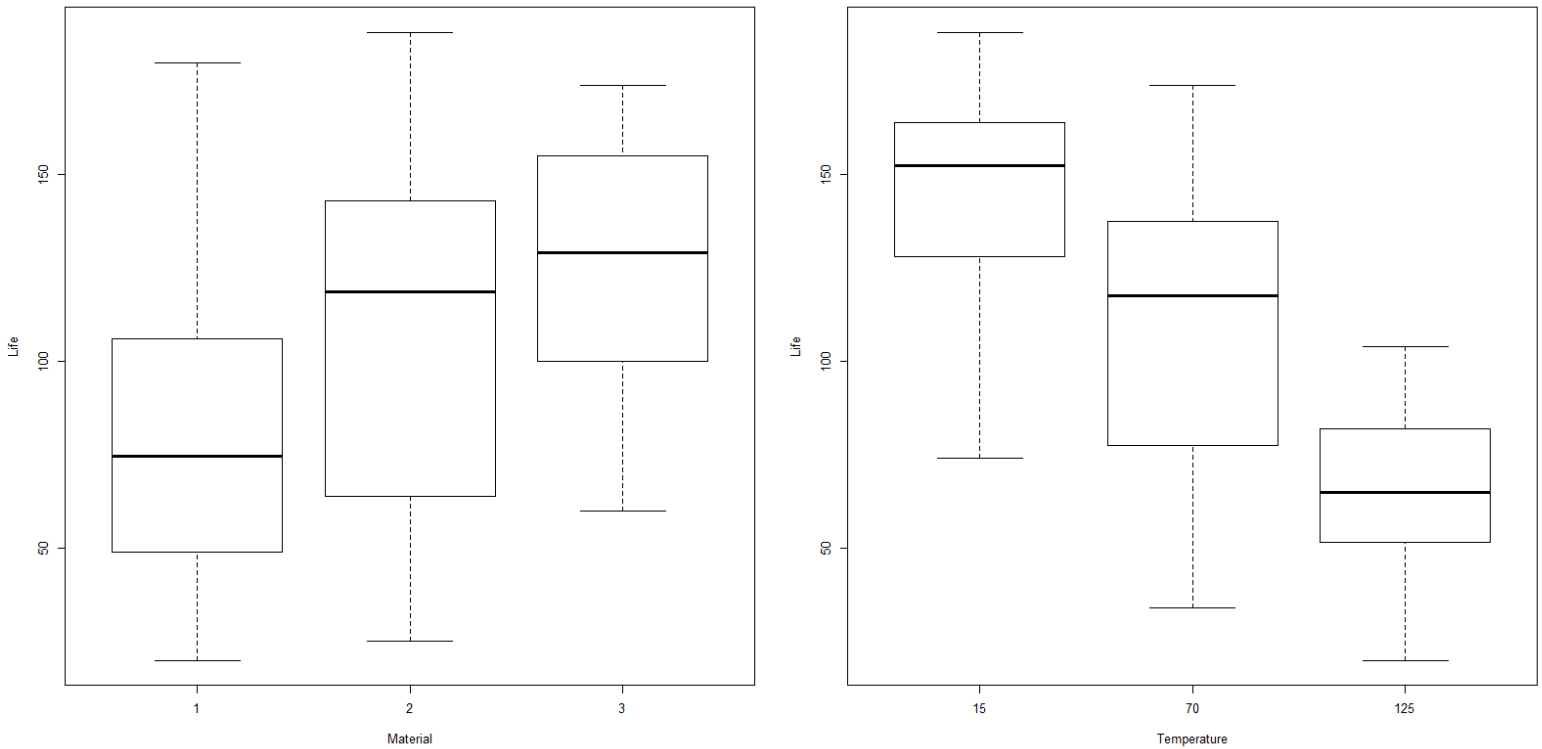
- Two factors: A = Material type; B = Temperature (quantitative variable)
- What effects do material type and temperature have on life?
- Is there a choice of material that would give uniformly long life regardless of temperature?

Life (in hours) Data for the Battery Design Example

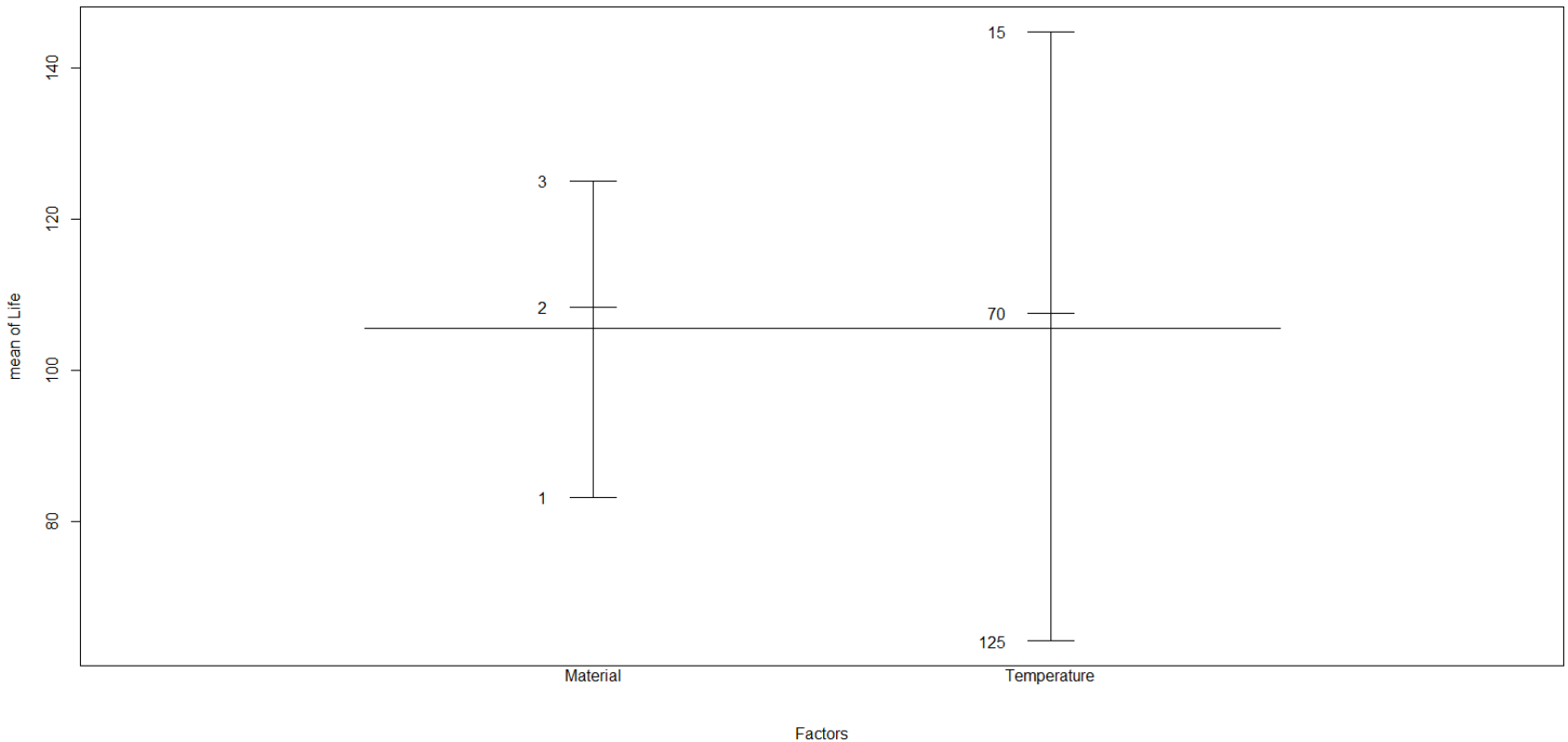
Material Type	Temperature (°F)					
	15		70		125	
1	130	155	34	40	20	70
	74	180	80	75	82	58
2	150	188	136	122	25	70
	159	126	106	115	58	45
3	138	110	174	120	96	104
	168	160	150	139	82	60

Since there are two factors at three levels, this design is sometimes called a 3^2 factorial design.

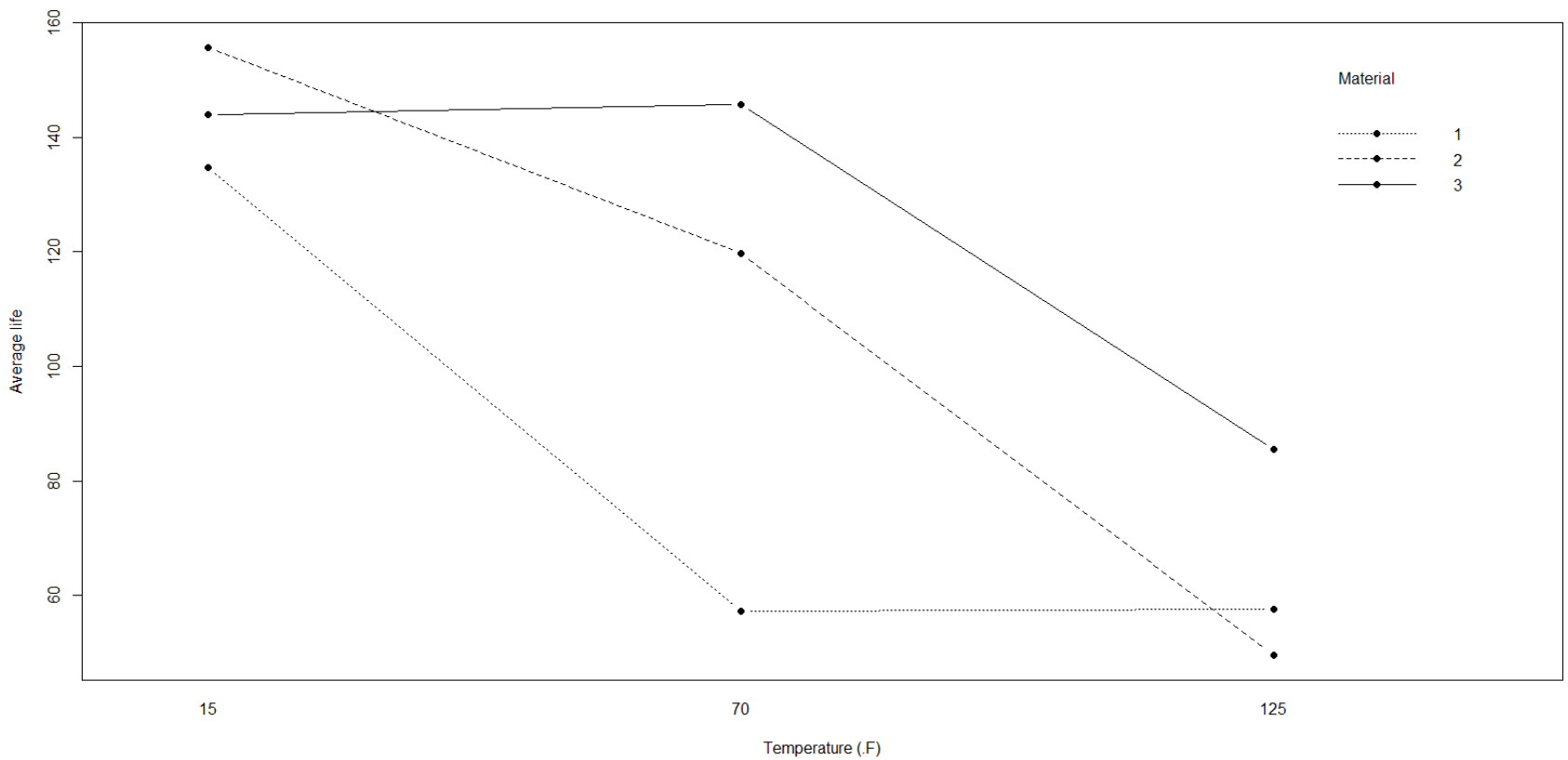
Box plot of data from the Battery Design Example



Effects plot of data from the Battery Design Example



Interaction plot of data from the Battery Design Example



ANOVA for Two-Factor Factorial Fixed-Effects Model

Effects model for the Two Factor - Fixed Factorial Experiment:

$$y_{ijk} = \mu_{ij} + \epsilon_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \epsilon_{ijk} \quad i = 1, \ldots, a; \; j = 1, \ldots, b; \; k = 1, \ldots, n.$$

where $(\tau\beta)_{ij}$ is the effect of the interaction between τ_i and β_j.

The Analysis of Variance Table for the Two-Factor Factorial, Fixed Effects Model

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
A treatments	SS_A	$a-1$	$MS_A = \frac{SS_A}{a-1}$	$F_0 = \frac{MS_A}{MS_E}$
B treatments	SS_B	$b-1$	$MS_B = \frac{SS_B}{b-1}$	$F_0 = \frac{MS_B}{MS_E}$
Interaction	SS_{AB}	$(a-1)(b-1)$	$MS_{AB} = \frac{SS_{AB}}{(a-1)(b-1)}$	$F_0 = \frac{MS_{AB}}{MS_E}$
Error	SS_E	$ab(n-1)$	$MS_E = \frac{SS_E}{ab(n-1)}$	
Total	SS_T	$abn-1$		

$$SS_E = SS_T - SS_{AB} - SS_A - SS_B = 77646.97 - 10683.72 - 39118.72 - 9613.78 = 18230.75$$

ANOVA for Battery Design Example

```
URL = ("https://raw.githubusercontent.com/francji1/01NAEX/"
       "main/data/")
bat = pd.read_csv(URL + "battery.txt", sep=r"\s+")
full = smf.ols("Life ~ C(Material) * C(Temperature)",
               data=bat).fit()
sm.stats.anova_lm(full, typ=2)
                              sum_sq  df       F     PR(>F)
C(Material)                10683.72   2   7.911    0.00198
C(Temperature)             39118.72   2  28.968   1.91e-07
C(Material):C(Temperature)  9613.78   4   3.560    0.01861
Residual                   18230.75  27
```

Both main effects and the interaction are significant at $\alpha = 0.05$: the effect of material depends on temperature.

`typ=2` (or `typ=1`, `typ=3`) gives the same table here, because the design is balanced.

ANOVA for Battery Design Example

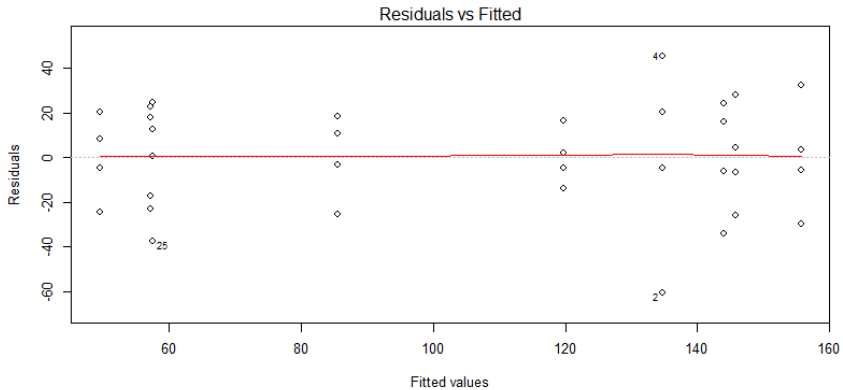
Additive model (no interaction) and the F-test comparing the two nested models:

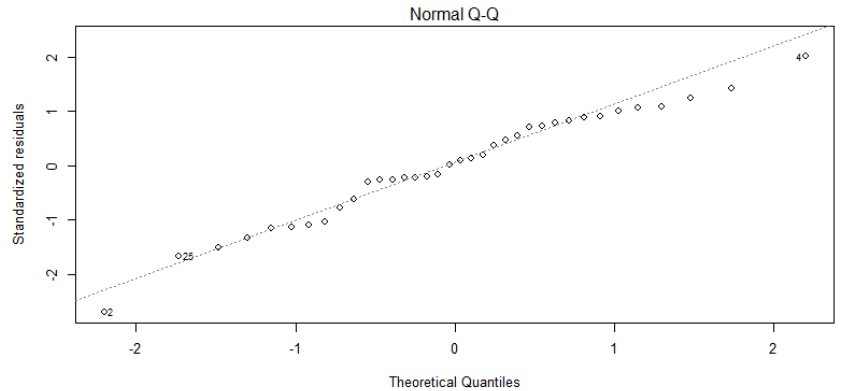
```
add = smf.ols("Life ~ C(Material) + C(Temperature)",
              data=bat).fit()
sm.stats.anova_lm(add, typ=2)
                    sum_sq  df        F     PR(>F)
C(Material)      10683.72   2    5.947    0.00651
C(Temperature)   39118.72   2   21.776   1.24e-06
Residual         27844.53  31

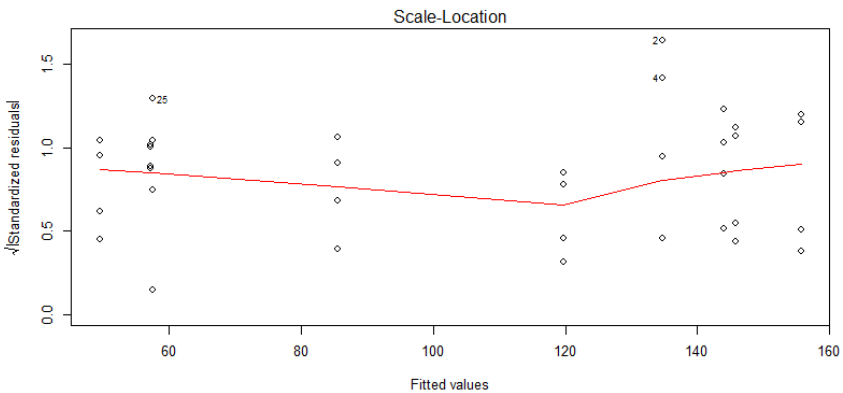
sm.stats.anova_lm(add, full)    # F-test for the interaction
   df_resid        ssr  df_diff   ss_diff      F    Pr(>F)
0      31.0  27844.53
1      27.0  18230.75      4.0   9613.78  3.560   0.01861
```

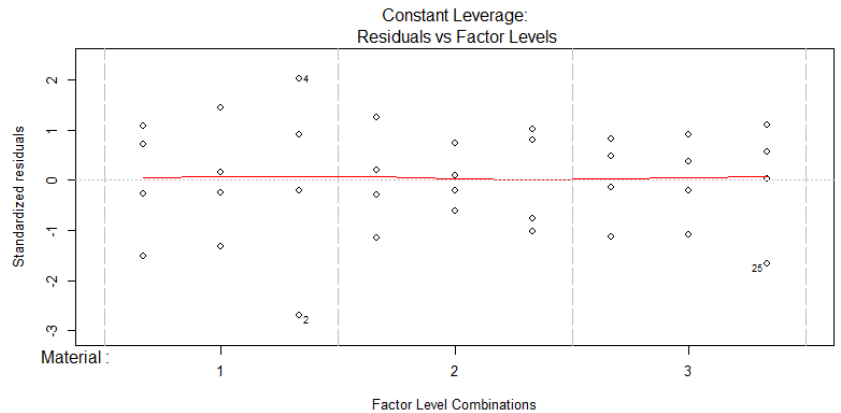
Dropping the interaction moves SS_{AB} into the error term; the comparison of the two models reproduces the interaction test of the full ANOVA table.

Model checking for Battery Design Example









Comparison between groups

Cell means and cell variances (Material $\times$ Temperature, $n = 4$ per cell):

```
bat.pivot_table(index="Material", columns="Temperature",
                values="Life", aggfunc="mean")
Temperature      15      70     125
Material
1            134.75   57.25    57.5
2            155.75  119.75    49.5
3            144.00  145.75    85.5

bat.pivot_table(index="Material", columns="Temperature",
                values="Life", aggfunc="var")
Temperature       15       70      125
Material
1            2056.92   556.92   721.00
2             656.25   160.25   371.00
3             674.67   508.25   371.67
```

Comparisons between the treatment group means

Tukey HSD for the main effects, with MS_E and df_E of the full two-factor model (12 observations per level):

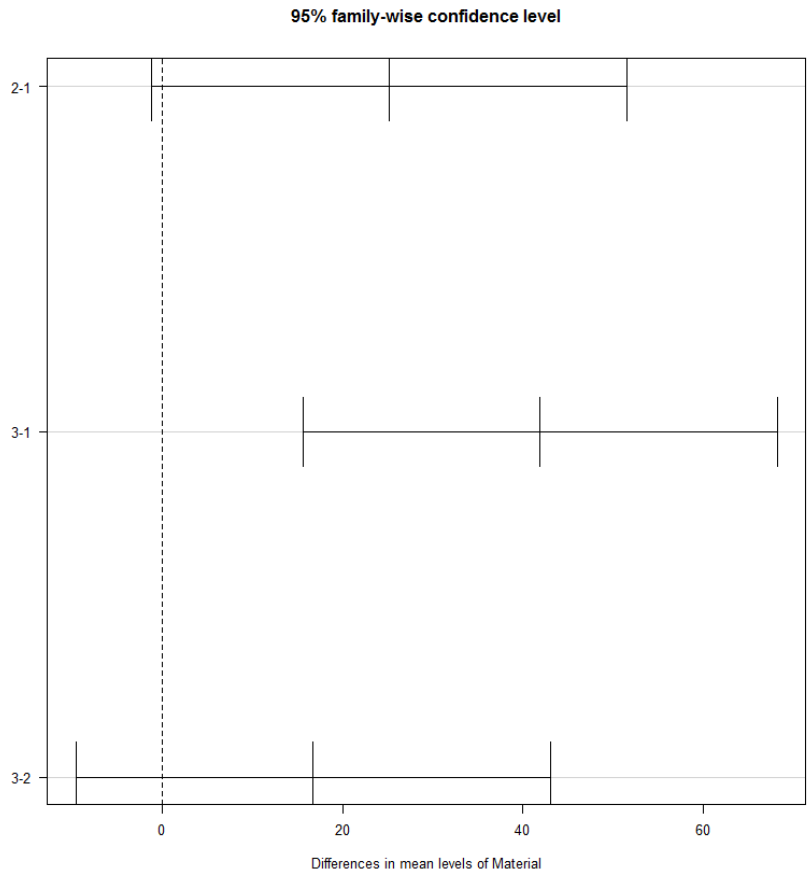
```
mse, dfe = full.mse_resid, full.df_resid        # 675.21, 27
q = stats.studentized_range.ppf(0.95, 3, dfe)   # 3.506
T = q * np.sqrt(mse / 12)                       # 26.30
```

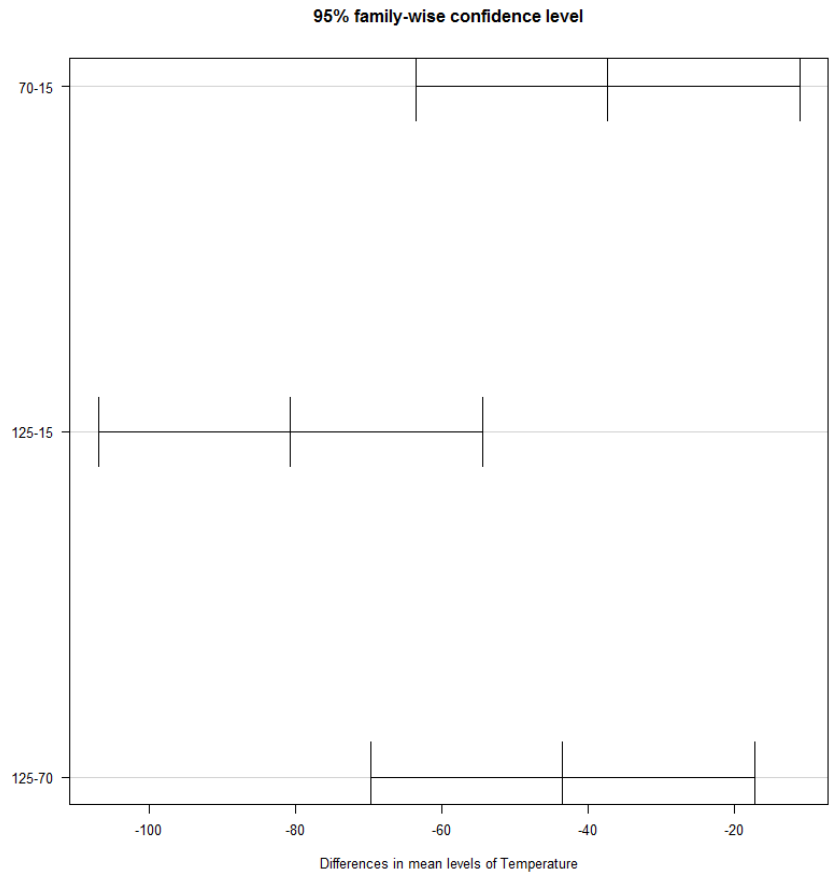
Material	diff	lwr	upr	p adj
2-1	25.17	-1.14	51.47	0.0628
3-1	41.92	15.61	68.22	0.0014
3-2	16.75	-9.55	43.05	0.2718
Temperature				
70-15	-37.25	-63.55	-10.95	0.0044
125-15	-80.67	-106.97	-54.36	1.0e-07
125-70	-43.42	-69.72	-17.11	0.0010

`pairwise_tukeyhsd(bat.Life, bat.Material)` would use a one-way model (temperature left in the error) and find no difference at all. Because the interaction is significant, the main-effect comparisons of material average over very different situations: compare materials at fixed temperatures.

Comparisons between the treatment group means

Visualization of Tukey HSD test:





Comparisons between the treatment group means

Because the interaction is significant, we compare the materials at one level of temperature, say 70°F, using the pooled MS_E of the full model ($n = 4$ per cell):

```
sub = bat[bat.Temperature == 70]
m70 = sub.groupby("Material").Life.mean()  # 57.25 119.75 145.75
q = stats.studentized_range.ppf(0.95, 3, dfe)          # 3.506
T70 = q * np.sqrt(mse / 4)                             # 45.56
m70[2] - m70[1], m70[3] - m70[1], m70[3] - m70[2]
# 62.50 88.50 26.00
```

$$T_{0.05} = q_{0.05}(3, 27)\sqrt{\frac{MS_E}{n}} = 3.506\sqrt{\frac{675.21}{4}} = 45.56$$

At 70°F the mean battery life is the same for material types 2 and 3 ($26.0 < 45.56$), and the mean life for material type 1 is significantly lower than for both types 2 and 3.

Simple Effects of Material at Each Temperature

The same comparison at every temperature (pooled MS_E, critical difference $T_{0.05} = 45.56$, $*$ = significant):

```
for t in [15, 70, 125]:
    sub = bat[bat.Temperature == t]
    m = sub.groupby("Material").Life.mean()
    print(t, m[2] - m[1], m[3] - m[1], m[3] - m[2])
```

```
Temperature      2-1       3-1       3-2
15             21.00      9.25    -11.75
70             62.50*    88.50*    26.00
125            -8.00     28.00     36.00
```

- The materials differ only at 70°F; this is the source of the interaction.
- Answer to the original question: material type 3 gives a long life at all temperatures (highest mean at 70°F and 125°F, not significantly lower at 15°F), so it is the most robust choice.

One Observation per Cell - Tukey's Test for Nonadditivity

Montgomery, Section 5.3.7

With $n = 1$ there are no df for pure error: SS_{AB} and SS_E cannot be separated. If we assume a simple interaction $(\tau\beta)_{ij} = \gamma\,\tau_i\beta_j$, the residual SS splits into

$$SS_N = \frac{\left[\sum_i \sum_j y_{ij}\, y_{i\cdot}\, y_{\cdot j} - y_{\cdot\cdot}\left(SS_A + SS_B + \frac{y_{\cdot\cdot}^2}{ab}\right)\right]^2}{ab\, SS_A\, SS_B} \quad (1 \text{ df}),$$

$$SS_{Error} = SS_{Residual} - SS_N \quad ((a-1)(b-1) - 1 \text{ df}),$$

and $F_0 = SS_N / \left(SS_{Error}/[(a-1)(b-1)-1]\right)$.

Equivalent regression trick: add the covariate $q_{ij} = \hat{\tau}_i\hat{\beta}_j$ to the additive model; its (sequential) SS is SS_N.

The same test checks the additivity assumption of the RCBD model (blocks $\times$ treatments). For the hardness data: $SS_N = 0.0041$, $F_0 = 0.43$, $p = 0.53$, no evidence of nonadditivity.

Tukey's Test for Nonadditivity - Impurity Example

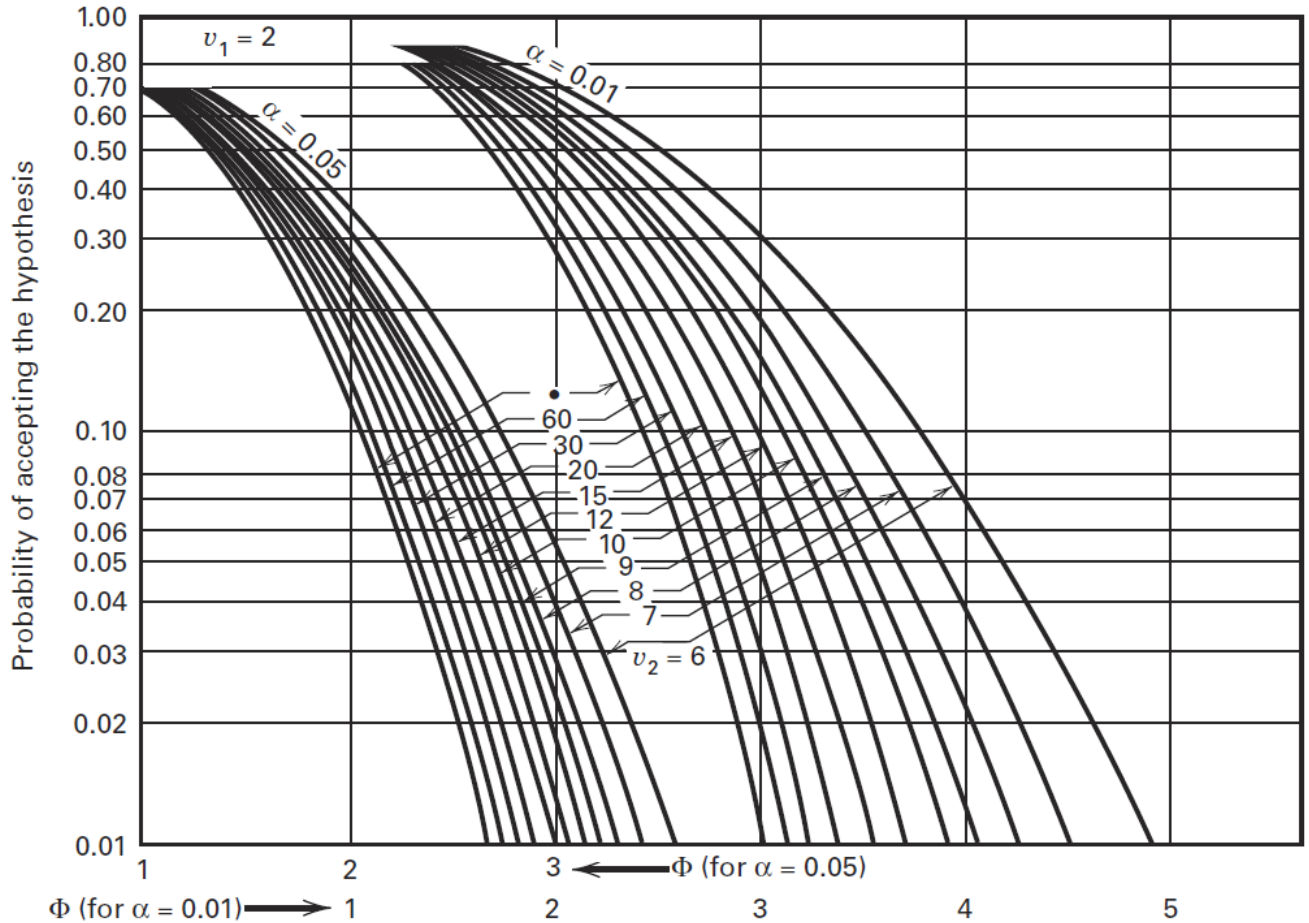
Example 5.2: impurity of a chemical product, temperature (3 levels) $\times$ pressure (5 levels), one replicate.

```
imp = pd.read_csv(URL + "impurity.txt", sep=r"\s+")
g = imp.N.mean()
ti = imp.groupby("Temperature").N.transform("mean") - g
bj = imp.groupby("Pressure").N.transform("mean") - g
imp["q"] = ti * bj
tuk = smf.ols("N ~ C(Temperature) + C(Pressure) + q",
              data=imp).fit()
sm.stats.anova_lm(tuk, typ=1)
                  df   sum_sq  mean_sq       F    PR(>F)
C(Temperature)   2.0  23.3333  11.6667  42.949    0.0001
C(Pressure)      4.0  11.6000   2.9000  10.676    0.0042
q                1.0   0.0985   0.0985   0.363    0.5660
Residual         7.0   1.9015   0.2716
```

No evidence of interaction ($p = 0.57$); the main effects of temperature and pressure are significant.

Sample Size Determination - Operating Characteristic Curve

Operating characteristic curves for the fixed-effects ANOVA with $\nu_1 = 2$ numerator df (Montgomery, Appendix Chart V). For factor A in the two-factor factorial: $\Phi^2 = \lambda_A / a$, $\nu_1 = a - 1$, $\nu_2 = ab(n-1)$.



Two-Treatment Factorial - What to Power

Model (balanced, n per cell):

$$y_{ijr} = \mu + \tau_i + \beta_j + \gamma_{ij} + \varepsilon_{ijr},$$

$$\mathrm{df}_A = a - 1,\ \mathrm{df}_B = b - 1,\ \mathrm{df}_{AB} = (a-1)(b-1),\ \mathrm{df}_E = ab(n-1).$$

Noncentrality parameters:

$$\lambda_A = \frac{nb}{\sigma^2}\sum_i (\mu_{i\cdot} - \bar{\mu})^2, \quad \lambda_B = \frac{na}{\sigma^2}\sum_j (\mu_{\cdot j} - \bar{\mu})^2,$$

$$\lambda_{AB} = \frac{n}{\sigma^2}\sum_{i,j} \left(\mu_{ij} - \mu_{i\cdot} - \mu_{\cdot j} + \bar{\mu}\right)^2.$$

OC parameters:

$$\Phi_A^2 = \lambda_A / a$$

$$\Phi_B^2 = \lambda_B / b$$

$$\Phi_{AB}^2 = \lambda_{AB} / \big((a-1)(b-1) + 1\big)$$

N: Set targets (D_A, D_B, D_{AB}), choose n to satisfy the strictest β requirement.

Detectable Sizes - Worst-Case Configurations

Main effect A (pairwise $\geq D_A$):

$$\sum_i (\mu_{i\cdot} - \bar{\mu})^2 \geq D_A^2/2 \;\Rightarrow\; \boxed{\Phi_A^2 = \frac{nb\,D_A^2}{2a\sigma^2}}.$$

Main effect B (pairwise $\geq D_B$):

$$\sum_j (\mu_{\cdot j} - \bar{\mu})^2 \geq D_B^2/2 \;\Rightarrow\; \boxed{\Phi_B^2 = \frac{na\,D_B^2}{2b\sigma^2}}.$$

Interaction $A \times B$ (diff-of-diffs $\geq D_{AB}$): Embed a single 2×2 pattern with entries $\pm c$ so that the interaction contrast equals D_{AB}; this gives SS $= D_{AB}^2/4$ and

$$\boxed{\Phi_{AB}^2 = \frac{n\,D_{AB}^2}{4\,[(a-1)(b-1)+1]\sigma^2}}.$$

N: These are least favorable (hardest to detect) alternatives.

Choosing n in Factorial - Per Effect, Take the Max

From OC tables/curves, obtain $\Phi^2_\star$ for each effect at (α, df).

Planner formulas

$$n_A = \frac{2a\,\hat{\sigma}^2}{b\,D_A^2}\,\Phi^2_{A,\star}, \qquad n_B = \frac{2b\,\hat{\sigma}^2}{a\,D_B^2}\,\Phi^2_{B,\star},$$

$$n_{AB} = \frac{4\,[(a-1)(b-1)+1]\hat{\sigma}^2}{D_{AB}^2}\,\Phi^2_{AB,\star}.$$

Final choice:

$$\boxed{n = \max\{n_A, n_B, n_{AB}\}} \quad \text{(round up)}.$$

N: If interaction is likely, n_{AB} often dominates. If not, size for A and B but still *test* $A \times B$ and interpret simple effects if needed.

Sample Size Determination for Battery Experiment

Detect a difference of $D = 40$ h between any two material means, $\sigma = 25$ h, $\alpha = 0.05$: $\Phi^2 = \dfrac{nbD^2}{2a\sigma^2}$, $\lambda = a\Phi^2$, $\nu_1 = a - 1 = 2$, $\nu_2 = ab(n-1)$.

```
a = b = 3; D, sigma = 40, 25
for n in [2, 3, 4, 5]:
    lam = n * b * D**2 / (2 * sigma**2)       # = a * Phi^2
    df1, df2 = a - 1, a * b * (n - 1)
    fcrit = stats.f.ppf(0.95, df1, df2)
    power = stats.ncf.sf(fcrit, df1, df2, lam)
```

```
n           2       3       4       5
Phi     1.600   1.960   2.263   2.530
lambda   7.68   11.52   15.36   19.20
df2         9      18      27      36
power   0.542   0.803   0.923   0.972
```

$n = 4$ replicates (the design as run) give power about 0.92; $n = 3$ would give only 0.80.

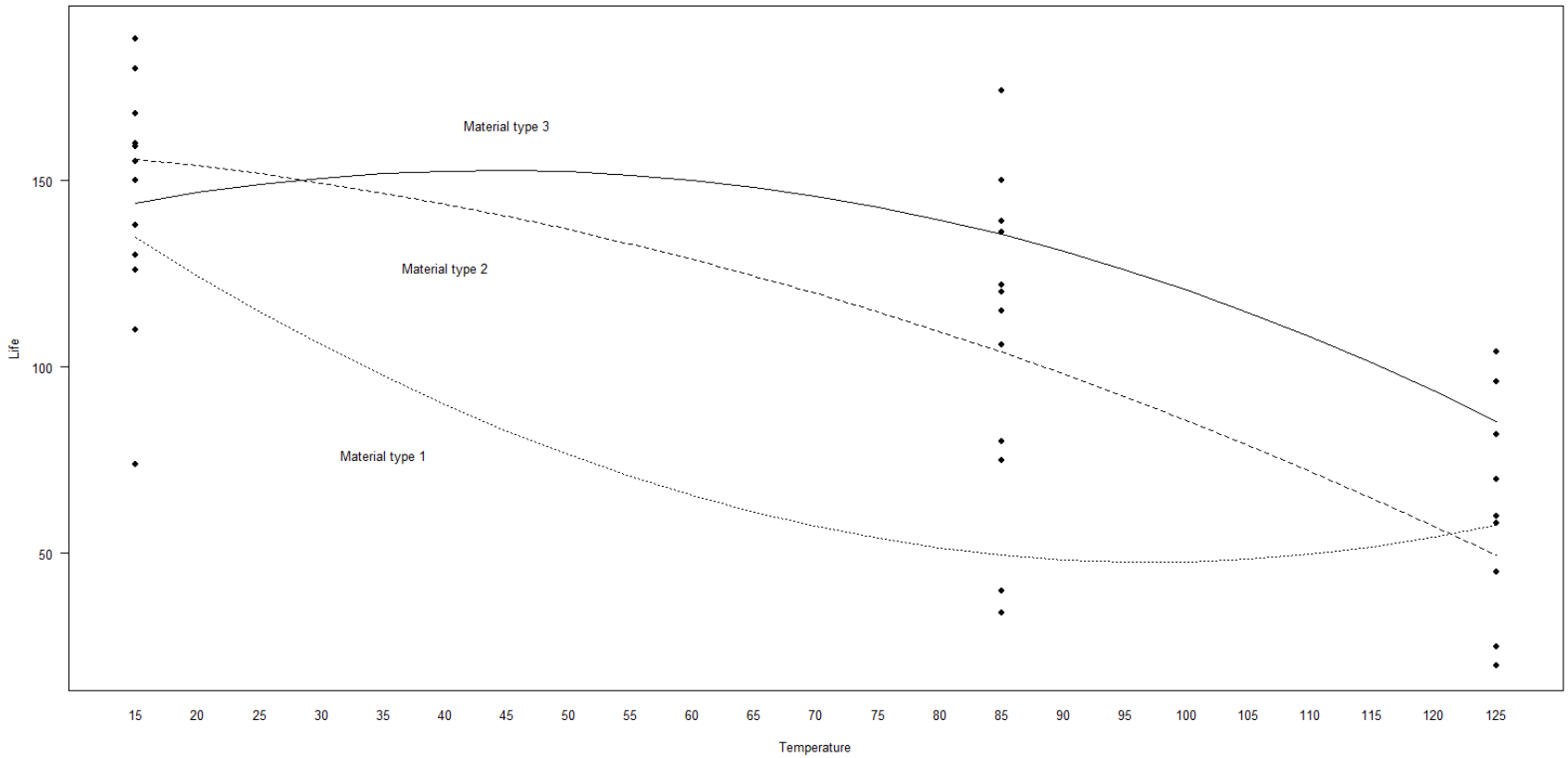
General Factorial Design - Response Curves

Temperature is quantitative: replace the factor by linear and quadratic terms (3 levels allow at most a quadratic) and let them interact with material.

```
bat["T"] = bat.Temperature.astype(float)
rc = smf.ols("Life ~ C(Material) + T + I(T**2)"
             " + C(Material):T + C(Material):I(T**2)",
             data=bat).fit()
sm.stats.anova_lm(rc, typ=1)
                       df     sum_sq       F      PR(>F)
C(Material)             2   10683.72   7.911     0.00198
T                       1   39042.67  57.823    3.53e-08
C(Material):T           2    2315.08   1.714     0.19911
I(T ** 2)               1      76.06   0.113     0.73975
C(Material):I(T ** 2)   2    7298.69   5.405     0.01061
Residual               27   18230.75
```

With A = material and B = temperature: B^2 and AB are not significant, whereas AB^2 is. Removing B^2 and AB would give a model that is not **hierarchical**, so we keep them. Note that $SS_B + SS_{B^2} = 39118.72$ and $SS_{AB} + SS_{AB^2} = 9613.78$, the temperature and interaction SS of the factorial ANOVA.

General Factorial Design - Response Curves



Fitted response curves (life vs temperature) for the three materials from the model with B, B^2, AB and AB^2. Material 3 gives the flattest curve, i.e. the life least sensitive to temperature.

General Factorial Design - More Than Two Factors

Montgomery, Section 5.4, Example 5.3

Factors A, B, C with a, b, c levels, n replicates; all $abcn$ runs in random order:

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk} + \varepsilon_{ijkl}.$$

- ANOVA with 7 effects; $n \geq 2$ is needed for an estimate of pure error ($abc(n-1)$ df).
- Interpret the highest significant interaction first, then the effects not involved in it.

Soft drink bottling (Example 5.3): response = deviation from the target fill height of a bottle.

- A = percent carbonation (10, 12, 14 %),
- B = operating pressure in the filler (25, 30 psi),
- C = line speed (200, 250 bottles per minute),
- $n = 2$ replicates, $3 \times 2 \times 2 \times 2 = 24$ runs.

General Factorial Design - Soft Drink Bottling

```
bot = pd.read_csv(URL + "bottling.txt", sep=r"\s+")
m3 = smf.ols("Deviation ~ C(Carbonation) * C(Pressure)"
             " * C(Speed)", data=bot).fit()
sm.stats.anova_lm(m3, typ=2)        # labels shortened
                       sum_sq  df         F      PR(>F)
Carbonation           252.750   2  178.412    1.19e-09
Pressure               45.375   1   64.059    3.74e-06
Speed                  22.042   1   31.118    1.20e-04
Carbonation:Pressure    5.250   2    3.706     0.0558
Carbonation:Speed       0.583   2    0.412     0.6715
Pressure:Speed          1.042   1    1.471     0.2486
Carb:Pressure:Speed     1.083   2    0.765     0.4869
Residual                8.500  12
```

- All three main effects are significant; the carbonation $\times$ pressure interaction is significant at 10 %.
- Mean deviation: carbonation $-0.50,\ 2.50,\ 7.38$; pressure $1.75,\ 4.50$; speed $2.17,\ 4.08$.
- Low carbonation and low pressure bring the fill height closest to the target; higher speed increases the deviation.

General Factorial Design - Response Surfaces

Montgomery, Section 5.5, Example 5.5

Tool life: A = cutting angle (15, 20, 25°), B = cutting speed (125, 150, 175 in/min), $n = 2$. Both factors are quantitative: fit a polynomial in both.

```
tool = pd.read_csv(URL + "toollife.txt", sep=r"\s+")
rs = smf.ols("Life ~ (Angle + I(Angle**2))"
             " * (Speed + I(Speed**2))", data=tool).fit()
sm.stats.anova_lm(rs, typ=1)
```

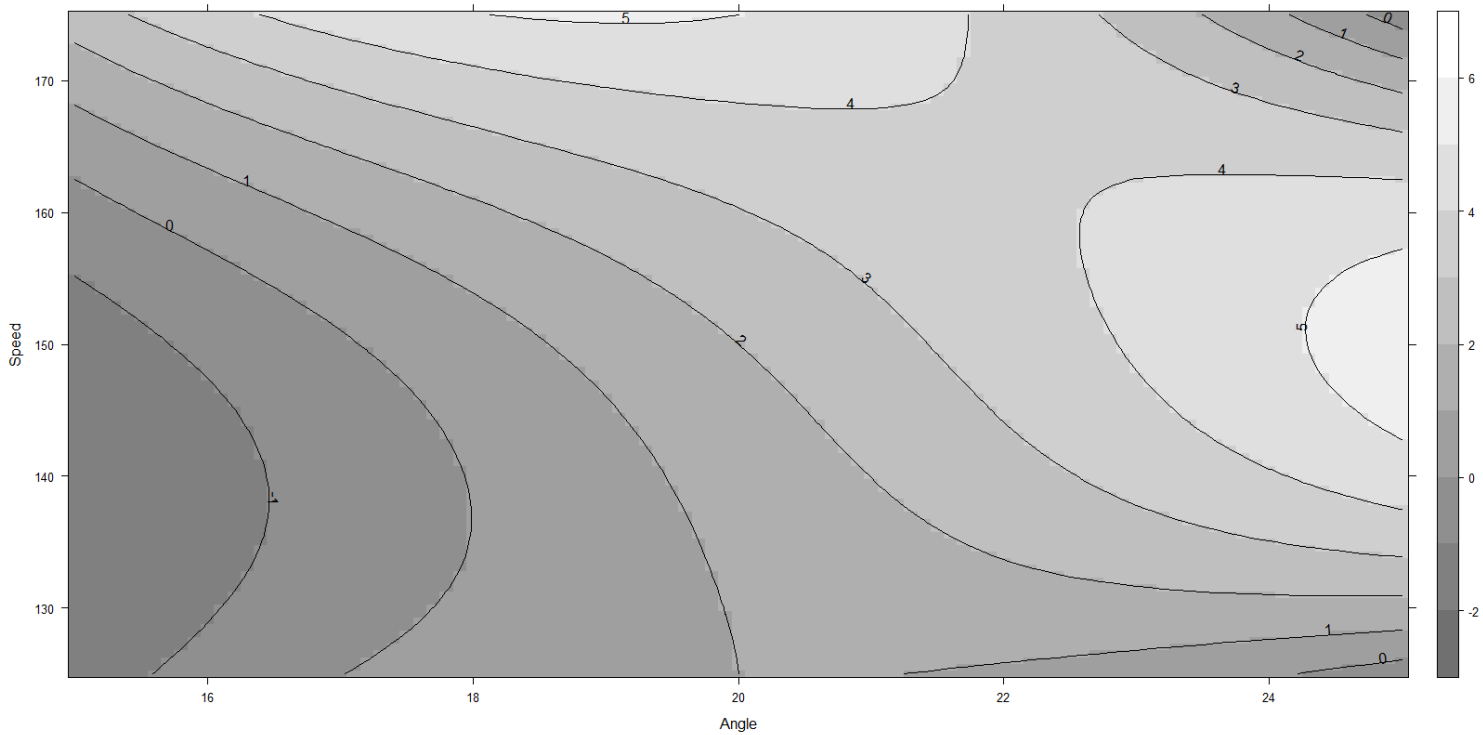
	df	sum_sq	F	PR(>F)
Angle	1	8.333	5.769	0.0398
I(Angle ** 2)	1	16.000	11.077	0.0088
Speed	1	21.333	14.769	0.0039
I(Speed ** 2)	1	4.000	2.769	0.1305
Angle:Speed	1	8.000	5.538	0.0431
Angle:I(Speed ** 2)	1	42.667	29.538	0.0004
I(Angle ** 2):Speed	1	2.667	1.846	0.2073
I(Angle ** 2):I(Speed ** 2)	1	8.000	5.538	0.0431
Residual	9	13.000		

The 8 terms reproduce the factorial ANOVA: $SS_A = 24.33$, $SS_B = 25.33$, $SS_{AB} = 61.33$ ($p = 0.0018$).

General Factorial Design - Response Surfaces

Fitted model (all 9 parameters, x_1 = angle, x_2 = speed):

$$\hat{y} = -1068 + 136.3x_1 - 4.08x_1^2 + 14.48x_2 - 0.0496x_2^2$$
$$- 1.864x_1x_2 + 0.0064x_1x_2^2 + 0.056x_1^2x_2 - 0.000192x_1^2x_2^2$$



The contour plot shows where the predicted tool life is longest; the optimum depends on both factors jointly (interaction), which a one-factor-at-a-time search would miss.

Blocking in a Factorial Design

Montgomery, Section 5.6, Example 5.6

If a complete replicate of the factorial can be run under homogeneous conditions (one day, one batch, one operator), each replicate is a **block**:

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \delta_k + \varepsilon_{ijk}, \qquad k = 1, \ldots, n \text{ (blocks)}.$$

- Runs are randomized **within** each block; no block $\times$ treatment interaction is assumed.
- The block effect uses $n - 1$ df of the error term.
- Random blocks: $E(MS_{Blocks}) = \sigma^2 + ab\,\sigma_\delta^2$, so $\hat{\sigma}_\delta^2 = (MS_{Blocks} - MS_E)/(ab)$.

Radar target detection: response = intensity level at which an operator detects a target.

- A = filter type (2 types), B = ground clutter (low, medium, high),
- operators differ in skill: each of 4 operators runs all 6 combinations in random order (operator = block),
- $2 \times 3 \times 4 = 24$ runs.

Blocking in a Factorial Design - Radar Example

```
radar = pd.read_csv(URL + "intensity.txt", sep=r"\s+")
blk = smf.ols("Intensity ~ C(Operator) + C(Filter)"
              " * C(Ground)", data=radar).fit()
sm.stats.anova_lm(blk, typ=2)
                          sum_sq  df        F      PR(>F)
C(Operator)              402.167   3  12.089    0.000277
C(Filter)               1066.667   1  96.192    6.45e-08
C(Ground)                335.583   2  15.132    0.000253
C(Filter):C(Ground)       77.083   2   3.476      0.0575
Residual                 166.333  15
```

- Filter type and ground clutter are significant, the interaction only at 10 %; operators differ strongly ($\hat{\sigma}^2_\delta = (134.06 - 11.09)/6 = 20.5$).
- Without the operator blocks the residual SS would be 568.5 with 18 df ($MS_E = 31.6$ instead of 11.1) and the ground clutter effect would have $p = 0.015$ instead of 0.0003.
- The same idea will matter in Team Project 1: each team member who measures is a natural block.

Today Exercise & Next Lecture

Today Exercise (Ex03):

- Use the prepared Jupyter notebook code/01NAEX_Ex03.ipynb (Python, Google Colab).
- Lecture examples in Python (hardness RCBD, battery life, radar), then in-class Problems 4.3, 5.4–5.5 and 5.15 from Montgomery.
- Assignment: Problems 4.21, 5.8 with 5.23, 5.18 and 5.20. Submit by pull request into `HW/` by Sunday 18 October 2026.

Next Lecture: Latin Squares and Incomplete Blocks

- Latin square and Graeco-Latin square designs (two and three blocking factors).
- Balanced incomplete block designs (BIBD).
- Team Project 1 will be assigned at the end of Lecture 4 (19 October).